RS overlap simplicity: independent mathematical review

ACCEPT 11 October 2026 UTC

Decision and exact accepted theorem

ACCEPT. The all-size argument is valid. No mathematical repair is needed. The complex critical-parameter annulus, Galois passage, residual nonvanishing, small-order certificates, and rational large-order estimate together prove the stated two subfamilies. The acceptance concerns mathematical validity; historical priority is unresolved.

Let $A \geq 1$ be an integer, d odd, and ω any primitive dth root of unity in characteristic zero. Assume either $b = 1$, $d \geq 3$, or $b = 3$, $d \geq 5$. With the accepted RS normalization,

$$F(z) = 1 + z^d, \qquad G(z) = H_b(z;\omega) = (z+1)U(z), \qquad g = U(-1),$$

$$W = \frac{(\omega;\omega)_b}{(\omega^2;\omega^2)_{(d+b)/2}}, \quad \tau = \frac{dW}{g}, \quad J_{A+1}(x;\tau) = x\,\mathrm{He}_A(x) - A\tau\,\mathrm{He}_{A-1}(x),$$

we have

$$\boxed{g \neq 0, \qquad \mathrm{Disc}(U)\,\mathrm{Res}(F,U) \neq 0, \qquad \mathrm{Disc}_x(J_{A+1}(x;\tau)) \neq 0.}$$

Here He_j is monic probabilists' Hermite; all discriminants are ordinary polynomial discriminants; $\mathrm{Disc}(1) = \mathrm{Res}(F,1) = 1$.

Thus the accepted manuscript's Section 3.2 simplicity and separation guards are automatic *within these two subfamilies*. Its local expansion and complete leading ordinary-discriminant coefficient then apply unchanged. This review does not alter the manuscript or remove any guard elsewhere. There is no assertion for $b \geq 5$, even d, $A = 0$, nonprimitive roots, or uniform growing-degree estimates.

Review method and independence

The critical-parameter matrix argument was reconstructed independently before reading the candidate proof. The remaining arguments were checked from definitions and exact algebra. The audit uses $y = x^2$ to check the Jacobi matrix, rather than the candidate's $y = x^2/2$ normalization. Independent exact controls use the complementary odd cyclotomic product, rather than the candidate's inverse-even product. The original uploaded programs and the candidate control program were treated as inert source; only the new review control program was executed. The all-A and all-d claims rest on the proof below, not finite-grid extrapolation.

The auditee's exact source and certificate identities, the frozen accepted manuscript, and the corrected prior-work review are recorded in `SOURCE_PINS.json`. The package manifest hashes each review payload. The separate final Library receipt checks the delivered PDF and ZIP against local bytes.

1. The critical-parameter annulus survives every exception

For $A = 1, 2$,

$$J_2 = x^2 - \tau, \qquad J_3 = x(x^2 - 1 - 2\tau).$$

Their multiple-root parameters are respectively 0 and $-1/2$. More generally, when A is odd, $J_{A+1}(0) = -A\tau \operatorname{He}_{A-1}(0)$; when A is even, $J'_{A+1}(0) = (1 + A\tau) \operatorname{He}_A(0)$. Therefore a nonreal critical τ has $A \geq 3$, and every multiple root x is nonzero.

Consecutive Hermite polynomials have no common root by their recurrence. At a common zero of J, J', both relevant Hermite factors are nonzero. Differentiation and elimination give

$$J' = (1 + A\tau) \operatorname{He}_A + Ax(1 - \tau) \operatorname{He}_{A-1}, \qquad y = x^2 = \frac{\tau(A\tau + 1)}{\tau - 1}.$$

No denominator vanishes because $\tau \notin \mathbb{R}$ and $x \neq 0$.

Write $A+1 = 2m+s$ with $s \in \{0, 1\}$ and $\alpha = s - 1/2$. The quadratic Hermite--Laguerre identity gives $J(x) = x^s \det(x^2 I - M)$, where M is complex symmetric tridiagonal, its fixed off-diagonal entries are

$$2\sqrt{j(j + \alpha)} > 0 \quad (1 \leq j < m),$$

and the diagonal entries before perturbation are $4j + 2\alpha + 2$ for $0 \leq j < m$. Add $A(\tau - 1)$ to the last diagonal. In particular the final entry becomes $A - 1 + A\tau$. The factor A, or $A/2$ in the candidate's half-scaled variable, is correct.

The eigenspace at any eigenvalue is one-dimensional, because its last coordinate determines the preceding ones and cannot be zero. A repeated eigenvalue consequently has a Jordan chain $(M - yI)w = v$, $(M - yI)v = 0$. Complex symmetry, using the bilinear transpose rather than adjoint, yields

$$v^T v = v^T (M - yI)w = ((M - yI)v)^T w = 0.$$

Hence $|v_m|^2 \leq \sum_{j<m} |v_j|^2$. Separately the Hermitian inner product gives

$$\frac{\operatorname{Im} y}{\operatorname{Im} \tau} = At, \qquad t = \frac{|v_m|^2}{\|v\|^2}, \qquad 0 < t \leq \frac{1}{2}.$$

Taking the imaginary part of the eliminated formula, with $\rho = |\tau - 1|^2$, gives

$$At = A - \frac{A + 1}{\rho}, \qquad \boxed{1 + \frac{1}{A} < \rho \leq 2 + \frac{2}{A} \leq \frac{8}{3}.}$$

Both inequalities have the correct strictness. In particular the upper endpoint cannot be replaced by a strict bound using this argument. No real-root theorem or diagonalizability assumption has been imported.

2. Arithmetic and residual guards

Put $d = 2h + 1$ and $E = \prod_{j=1}^{h}(1 - \omega^{2j})$. The negative even residues complement the positive even residues, also for composite d. Thus

$$E\overline{E} = d, \quad E^2 = (-1)^h d\omega^{h(h+1)}, \quad \tau_1 = d/E = \prod_{j=1}^{h}(1 - \omega^{2j-1}).$$

Since $4h(h + 1) = d^2 - 1$, its exponent is coprime to d. Therefore τ_1^2 is nonreal for odd $d \geq 3$, and so is τ_1, in every primitive embedding. For $b = 1$, $U = g = 1$.

For $b = 3$, exact cancellation gives

$$U(z) = z^2 + (\omega + \omega^2)z + 1, \quad g = (1 - \omega)(2 + \omega), \quad \tau_3 = \tau_1 \frac{1 + \omega}{2 + \omega}.$$

The coefficient $\omega + \omega^2$ is nonreal for odd order $d \geq 5$: its imaginary part is $\sin\theta(1 + 2\cos\theta)$, whose root-of-unity exceptions have orders $1, 2, 3$. Consequently $g \neq 0$, $\mathrm{Disc}(U) \neq 0$, and U has no unit-modulus root. Every root of F has unit modulus, proving $\mathrm{Res}(F, U) \neq 0$.

If τ_3 were real, conjugation and fourth powers would force

$$0 = (\omega - 1)(\omega^2 + \omega + 1)(16\omega^2 + 31\omega + 16).$$

The first two factors are impossible for $d \geq 5$. The last would give the algebraic integer $\omega + \omega^{-1} = -31/16$, a noninteger rational, also impossible. Thus τ_3 is nonreal in every embedding. All denominators in W are nonzero because $(d + b)/2 < d$ and ω^2 is primitive of odd order d.

3. Galois exclusion and the exact small orders

The discriminant $D_A(T) = \mathrm{Disc}_x J_{A+1}(x; T)$ lies in $\mathbb{Q}[T]$. If $D_A(\tau) = 0$, then $D_A(\sigma\tau) = 0$ for every cyclotomic automorphism σ. Those automorphisms commute with complex conjugation, so

$$\sigma\big((\tau - 1)(\overline{\tau} - 1)\big) = |\sigma\tau - 1|^2.$$

Therefore one conjugate norm outside the annulus suffices. It is unnecessary that every embedding satisfy the same outer estimate.

Exact quotient-ring resultants confirm the following polynomials for ρ; the norm is the monic polynomial raised to $\varphi(d)/\deg P$:

$$\begin{aligned}
(b, d) = (1, 3), (3, 7): &\quad r - 1, \\
(1, 5): &\quad r^2 - 7r + 11, \\
(3, 5): &\quad 11r^2 - 7r + 1, \\
(3, 9): &\quad 19r^3 - 129r^2 + 282r - 199, \\
(3, 11): &\quad 683r^5 - 8684r^4 + 42063r^3 \\
&\qquad - 97250r^2 + 107618r - 45761.
\end{aligned}$$

For $(1,3)$ and $(3,7)$, $\rho = 1$ violates the strict inner bound. For $(1,5)$, a root lies in $(4,5)$ because the polynomial values at $4,5$ are $-1,1$. For $(3,5)$, a root lies in $(0,1/4)$, since the values are $1,-1/16$. The latter already suffices, although both roots are below 1.

For $(3,9)$ the values at $8/3$ and 4 are $-109/27$ and 81, giving a conjugate above $8/3$. For $(3,11)$ the values at $4,5$ are $-2969, 25829$, again giving an outer conjugate. These are exact sign arguments with exact norm identities; irreducibility of the displayed polynomial need not be presumed.

For $b = 1$ and $d \geq 7$, every embedding satisfies

$$\rho \geq (\sqrt{d}-1)^2 \geq (\sqrt{7}-1)^2 > 8/3,$$

using $\sqrt{7} < 8/3$. This proves all remaining $b = 1$ cases.

4. The bound for every odd $d \geq 13$ is rigorous

For $b = 3$, select $\omega = e^{2\pi i/d}$, let $v = \pi/d$, $a = \pi/4 - v/4$, and $t = \tan v$. The signs of the sine factors in E fix the phase:

$$E = \sqrt{d}\, e^{ia}, \qquad \tau_3 = \frac{2\sqrt{d}\, e^{-ia}}{3-it},$$

$$\rho = 1 + \frac{4d}{9+t^2} - \frac{4\sqrt{d}(3\cos a + t\sin a)}{9+t^2}.$$

The bounds are exact. Since $157/50 < \pi < 22/7$, put $u = 22/91$ and $\ell = 471/650$. Then

$$0 < t < \frac{u}{1-u^2/2} = \frac{2002}{8039} < \frac{1}{4},$$

by $\sin v < v$ and $\cos v > 1 - v^2/2$. Also $a \geq 3\pi/13 > \ell$, so cosine monotonicity and its alternating Taylor bound imply

$$\cos a < 1 - \frac{\ell^2}{2} + \frac{\ell^4}{24} < \frac{3}{4}.$$

Finally $0 < a < \pi/4$ gives $\sin a < 1/\sqrt{2} < 3/4$. Bounding the positive and negative terms separately therefore gives

$$\rho > f(d) := 1 + \frac{64d}{145} - \frac{13\sqrt{d}}{12}.$$

For real $d \geq 13$, $f'(d) = 64/145 - 13/(24\sqrt{d}) > 64/145 - 13/72 > 0$. Moreover

$$f(13) > 1 + \frac{832}{145} - \frac{13}{12}\frac{361}{100} = \frac{8}{3} + \frac{1861}{11600} > \frac{8}{3}.$$

Thus one principal embedding violates the outer annulus at every larger odd order, and Galois conjugacy proves simplicity in every embedding. There is no unhandled odd d or A.

5. Controls, provenance, and integration boundary

The independent program `independent_exact_checks.py` records exact quotient-ring norm identities and their multiplicities, rational sign brackets, the reality-obstruction factorization, the residual nonreality identity, all rational inequalities, Hermite derivative identities, parity-Jacobi characteristic polynomials, and direct RS product specializations. It additionally checks finite actual discriminants for $1 \le A \le 12$ and odd $3 \le d \le 15$, with $b \in \{1, 3\}$ and $b < d$. All 249 independent exact checks passed. The output records every check. These finite checks corroborate normalization only; Sections 1--4 are the all-size proof.

No preexisting source program was executed. The author certificate's reported 86 checks was inspected as evidence, and the key identities were independently recomputed. The accepted manuscript PDF and LaTeX, corrected prior-work review, and final candidate source have separate byte identities in the review package. Hashes establish file identity, not mathematical truth.

Integration rule. This accepted extension may justify a clearly scoped corollary or subfamily annotation in the manuscript. Keep the general $\mathrm{Disc}(U)\,\mathrm{Res}(F, U)\,\mathrm{Disc}(J) \neq 0$ hypothesis for all other residual degrees. Do not infer a theorem for $b \ge 5$, even d, or a different normalization of τ. Do not change ranks or publication-priority status based on this proof review. The accepted all-order RS local-expansion proof and the finite-parameter certificate remain distinct earlier results.

Sources

1. Audited candidate: *Simplicity of two Rogers Szego overlap subfamilies*, 11 October 2026. Exact final source, author certificate, and sealed package identities are in `SOURCE_PINS.json`.

2. Accepted source: *Local ordinary discriminants of Rogers Szego polynomials at odd roots of unity*, version 3, Section 3.2, equations (5)--(10).

   LaTeX SHA-256:
   `57bc17168b7b11efe7a5bdc7190473267`
   `b0c08fcd4f2bb5b9c59f04f1cbfa282`.

   PDF SHA-256:
   `47da02758de282ff76f4ae54dd5d07a6b`
   `583dd15f49bb8af39763242f43f805e`.

3. *Independent review of the Rogers Szego prior work audit*, 11 October 2026, repaired F8 comparison. The quasi-Laguerre family interface is retained; no real-coefficient simplicity theorem is imported.

4. NIST Digital Library of Mathematical Functions, Section 18.7, especially equations 18.7.11 and 18.7.19--20, and Section 18.9. These record the classical normalization, quadratic identities, recurrence, and derivative relations. Accessed 11 October 2026.

Final disposition: ACCEPT for the exact two-subfamily theorem above. The original manuscript and its general guards remain untouched by this review. No Sheet or repository write was performed.